

EASY

1. Who developed Python Programming Language?

- a) Wick van Rossum
- b) Rasmus Lerdorf
- c) Guido van Rossum
- d) Niene Stom

2. Which type of Programming does Python support?

- a) object-oriented programming
- b) structured programming
- c) functional programming
- d) all of the mentioned

3. Is Python case-sensitive when dealing with identifiers?

- a) no
- b) yes
- c) machine dependent
- d) none of the mentioned

4. Which of the following is the correct extension of the Python file?

- a) .python
- b) .pl
- c) .py
- d) .p

5. Is Python code compiled or interpreted?

- a) Python code is both compiled and interpreted
- b) Python code is neither compiled nor interpreted
- c) Python code is only compiled
- d) Python code is only interpreted

6. All keywords in Python are in _____

- a) Capitalized
- b) lower case
- c) UPPER CASE
- d) None of the mentioned

8. Which of the following is used to define a block of code in Python language?

- a) Indentation
- b) Key
- c) Brackets
- d) All of the mentioned

9. Which keyword is used for function in Python language?

- a) Function
- b) def
- c) Fun

d) Define

10. Which of the following characters is used to give single-line comments in Python?

- a) //
- b) #
- c) !
- d) /*

16. What does pip stand for Python?

- a) Pip Installs Python
- b) Pip Installs Packages
- c) Preferred Installer Program
- d) All of the mentioned

17. Which of the following is true for variable names in Python?

- a) underscore and ampersand are the only two special characters allowed
- b) unlimited length
- c) all private members must have leading and trailing underscores
- d) none of the mentioned

19. Which of the following is the truncation division operator in Python?

- a) |
- b) //
- c) /
- d) %

25. Which of the following is not a core data type in Python programming?

- a) Tuples
- b) Lists
- c) Class
- d) Dictionary

26. What will be the output of the following Python expression if x=56.236?

```
print("%.2f"%x)
```

- a) 56.236
- b) 56.23
- c) 56.0000
- d) 56.24

28. What will be the output of the following Python function?

```
len(["hello", 2, 4, 6])
```

- a) Error
- b) 6
- c) 4

d) 3

33. Which function is called when the following Python program is executed?

```
f = foo()
format(f)
```

- a) str()
- b) format()
- c) __str__()
- d) __format__()

34. Which one of the following is not a keyword in Python language?

- a) pass
- b) eval
- c) assert
- d) nonlocal

35. What will be the output of the following Python code?

```
1. class tester:
2.     def __init__(self, id):
3.         self.id = str(id)
4.         id="224"
5.
6. >>>temp = tester(12)
7. >>>print(temp.id)
```

- a) 12
- b) 224
- c) None
- d) Error

37. Which module in the python standard library parses options received from the command line?

- a) getarg
- b) getopt
- c) main
- d) os

38. What will be the output of the following Python program?

```
z=set('abc')
z.add('san')
z.update(set(['p', 'q']))
z
```

- a) {'a', 'c', 'c', 'p', 'q', 's', 'a', 'n'}
- b) {'abc', 'p', 'q', 'san'}
- c) {'a', 'b', 'c', 'p', 'q', 'san'}
- d) {'a', 'b', 'c', ['p', 'q'], 'san'}

41. Which of the following statements is used to create an empty set in Python?

- a) ()
- b) []
- c) { }
- d) set()

43. To add a new element to a list we use which Python command?

- a) list1.addEnd(5)
- b) list1.addLast(5)
- c) list1.append(5)
- d) list1.add(5)

45. What will be the output of the following Python code?

```
1. >>>list1 = [1, 3]
2. >>>list2 = list1
3. >>>list1[0] = 4
4. >>>print(list2)
```

- a) [1, 4]
- b) [1, 3, 4]
- c) [4, 3]
- d) [1, 3]

47. Which of the following Python statements will result in the output: 6?

```
A = [[1, 2, 3],
      [4, 5, 6],
      [7, 8, 9]]
```

- a) A[2][1]
- b) A[1][2]
- c) A[3][2]
- d) A[2][3]

48. What is the maximum possible length of an identifier in Python?

- a) 79 characters
- b) 31 characters
- c) 63 characters
- d) none of the mentioned

50. What will be the output of the following Python code?

```
x = 'abcd'
for i in range(len(x)):
    print(i)
```

- a) error
- b) 1 2 3 4
- c) a b c d

d) 0 1 2 3

51. What are the two main types of functions in Python?

- a) System function
- b) Custom function
- c) Built-in function & User defined function
- d) User function

52. What will be the output of the following Python program?

```
def addItem(listParam):  
    listParam += [1]  
  
mylist = [1, 2, 3, 4]  
addItem(mylist)  
print(len(mylist))
```

a) 5

b) 8

c) 2

d) 1

53. Which of the following is a Python tuple?

- a) {1, 2, 3}
- b) {}
- c) [1, 2, 3]
- d) (1, 2, 3)

54. What will be the output of the following Python code snippet?

```
z=set('abc$de')  
'a' in z
```

a) Error

b) True

c) False

d) No output

55. What will be the output of the following Python expression?

```
round(4.576)
```

a) 4

b) 4.6

c) 5

d) 4.5

56. Which of the following is a feature of Python DocString?

- a) In Python all functions should have a docstring
- b) Docstrings can be accessed by the `__doc__` attribute on objects
- c) It provides a convenient way of associating documentation with Python modules, functions, classes, and methods
- d) All of the mentioned

58. What is output of `print(math.pow(3, 2))`?

- a) 9.0
- b) None
- c) 9
- d) None of the mentioned

59. Which of the following is the use of `id()` function in python?

- a) Every object in Python doesn't have a unique id
- b) In Python `id` function returns the identity of the object
- c) None of the mentioned
- d) All of the mentioned

1. Role: Junior Software Developer at a Tech Company

You are tasked with writing a Python script to store the details of employees in a company. The details include their names, IDs, and departments. Which data structure is most appropriate for storing this information?

- A. List
- B. Set
- C. Dictionary
- D. Tuple

Correct Answer: C

2. Role: Game Developer Designing an Adventure Game

You need to create a loop that continuously checks if a player has lost all their lives. If the player loses all lives, the game ends. Which control flow structure will you use for this scenario?

- A. for loop with break statement
- B. while loop with break statement
- C. if-else inside a for loop
- D. pass statement inside a while loop

Correct Answer: B

3. Role: Data Scientist Analyzing Stock Market Data

You are given a list of stock prices for a company over the last 10 days. You want to write a function that returns the maximum price in the list. How will you define the function in Python?

A. `def max_price():`
 `return max(prices)`

B.

`def max_price(prices):`
 `return max(prices)`

C.

`def max_price():`
 `max(prices)`

D.

`def max_price(prices):`
 `prices.max()`

Correct Answer: B

5. Role: Analyst Handling Date and Time for Reports

You need to generate reports with timestamps that include the current date and time. Which Python module will you import to handle date and time in your script?

- A. `os`
- B. `datetime`
- C. `math`
- D. `random`

Correct Answer: B

6. Role: Data Engineer Cleaning Financial Data

You are processing a large CSV file containing financial data. You want to remove leading and trailing spaces from all text columns before analyzing the data. Which string method will you use to clean the text data?

- A. `strip()`
- B. `replace()`
- C. `find()`
- D. `split()`

Correct Answer: A

8. Role: Software Engineer Optimizing Data Search in Large Files

You are tasked with searching for a specific item in a list of one million elements. Which Python algorithmic approach would be most efficient for such a task?

- A. Linear search
- B. Binary search
- C. Bubble sort
- D. Quick sort

Correct Answer: B

9. Role: Backend Developer Handling File Uploads

Your Python application needs to read a text file uploaded by a user and display its contents on the screen. Which of the following modes will you use to open the file for reading?

- A. 'r'
- B. 'w'
- C. 'a'
- D. 'x'

Correct Answer: A

10. Role: Data Scientist Cleaning Missing Data from a Dataset

You are working with a dataset that has several missing values (NaNs). You want to replace all NaNs with a specified value, say 0, before analysis. Which pandas function will you use for this?

- A. `fillna()`
- B. `dropna()`
- C. `replace_na()`
- D. `isnull()`

Correct Answer: A

11. Role: Data Analyst Using Regular Expressions for Data Validation

You are writing a Python script to validate email addresses. You want to use regular expressions to check if the email is in a valid format. Which module should you import to use regular expressions in Python?

- A. os
- B. re
- C. sys
- D. json

Correct Answer: B

13. Role: DevOps Engineer Automating Server Logs Monitoring

You need to process server logs stored in multiple text files. These logs contain timestamps, and you want to extract the date and time from each entry. Which Python feature is best for extracting specific patterns like timestamps?

- A. File handling
- B. Regular expressions
- C. List comprehensions
- D. OOP

Correct Answer: B

1. Role: Data Analyst Cleaning Sales Data

You are working with a sales dataset in pandas, where some columns have missing values. You want to drop all rows that contain any missing values. Which method will you use?

- A. `df.dropna()`
- B. `df.fillna()`
- C. `df.drop()`
- D. `df.replace()`

Correct Answer: A

7. Role: Data Scientist Analyzing Customer Demographics

You are working with a dataset that contains information about customers, including their ages. You want to calculate the average age of all customers. Which pandas method will you use to get this value?

- A. `df['age'].mean()`
- B. `df['age'].sum()`
- C. `df['age'].avg()`
- D. `df['age'].median()`

Correct Answer: A

10. Role: Data Engineer Detecting Duplicate Entries

You are cleaning a dataset and want to identify if there are any duplicate rows. Which pandas method can help you find these duplicate rows?

- A. `df.duplicated()`
- B. `df.drop_duplicates()`
- C. `df.find_duplicates()`
- D. `df.check_duplicates()`

Correct Answer: A

12. Role: Data Analyst Renaming Columns for Consistency

You are working with a dataset where the column names are inconsistent. You want to rename the columns `Product ID` and `Customer ID` to `product_id` and `customer_id`, respectively. Which pandas method will you use?

- A. `df.rename(columns={'Product ID': 'product_id', 'Customer ID': 'customer_id'})`
- B. `df.replace(columns={'Product ID': 'product_id', 'Customer ID': 'customer_id'})`
- C. `df.columns = ['product_id', 'customer_id']`
- D. `df.update(columns={'Product ID': 'product_id', 'Customer ID': 'customer_id'})`

Correct Answer: A

Question 4: Merging DataFrames

You have two DataFrames:

`df1:`

Product	Price
Laptop	1000
Phone	500

`df2:`

Product	Quantity
Laptop	50
Phone	100

Question:

Which of the following commands will merge df1 and df2 on the "Product" column?

- A) `pd.merge(df1, df2, on='Product')`
- B) `pd.concat([df1, df2], axis=1)`
- C) `df1.join(df2, on='Product')`
- D) Both A and B

Question 5: Handling Missing Data

You have the following DataFrame with missing values:

Name	Age	Salary
Alice	25	50000
Bob	NaN	70000
Charlie	35	NaN
David	40	80000

Question:

Which command will fill the missing values in the "Salary" column with the mean salary?

- A) `df['Salary'].fillna(df['Salary'].mean(), inplace=True)`
- B) `df['Salary'].replace(np.nan, df['Salary'].mean(), inplace=True)`
- C) `df['Salary'].fill(df['Salary'].mean())`
- D) `df['Salary'] = df['Salary'].apply(lambda x: x if x != np.nan else df['Salary'].mean())`

Question 8: Pivot Tables

Consider the following DataFrame:

Name	Year	Sales
Alice	2022	500
Bob	2022	300
Charlie	2023	700

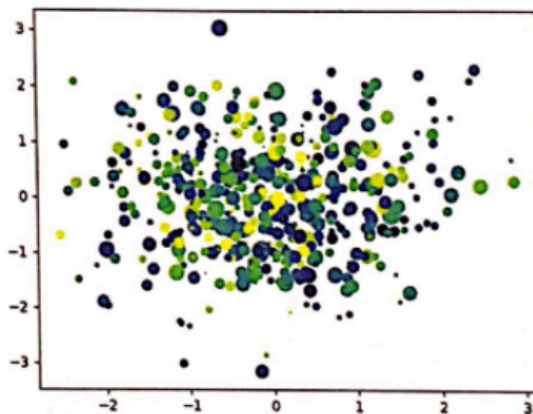
Name	Year	Sales
David	2023	800

Question:

Which command will create a pivot table showing the total sales per year?

- A) `df.pivot_table(values='Sales', index='Year', aggfunc='sum')`
- B) `pd.pivot_table(df, values='Sales', index='Year', aggfunc='sum')`
- C) `df.pivot('Year', values='Sales', aggfunc='sum')`
- D) Both A and B

9. Identify the following chart's type



- Scatter Chart
- Bubble Chart
- Pie Chart
- Frequency Polygon

Show Answer

a. Scatter Chart

20. in the given chart, box surrounded with red border is called

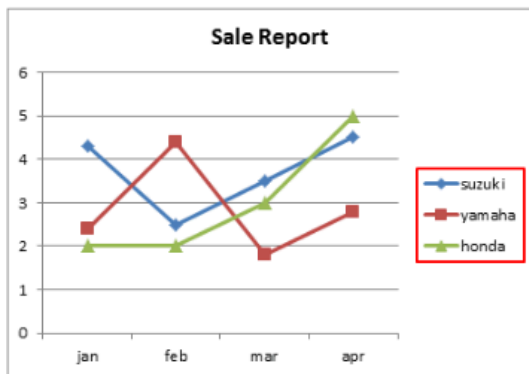


- a. Label
- b. X label
- c. Y label
- d. Legend

Show Answer

b. X label

3. In the given chart, box surrounded with red border is called



- a. Data series
- b. Chart Title
- c. Markers
- d. Legend

Show Answer

d. Legend

Question:

```
import pandas as pd
data = [['Anuj', 21], ['Rama', 25], ['Kapil', 22]]
df = pd.DataFrame(data, columns=['Name', 'Age'])
print(df)
```

A.

	Name	Age
0	Anuj	21
1	Rama	25
2	Kapil	22

B.

	Name	Age
0	Anuj	21
1	Kapil	22
2	Rama	25

C.

	Name	Age
0	Kapil	22
1	Rama	25
2	Anuj	21

D.

	Name	Age
0	Rama	25
1	Anuj	21
2	Kapil	22

Answer: A)

MEDIUM

11. What will be the output of the following Python code?

```
i = 1
while True:
    if i%3 == 0:
        break
    print(i)

    i += 1
```

- a) 1 2 3
- b) error
- c) 1 2**
- d) none of the mentioned

14. What is the order of precedence in python?

- a) Exponential, Parentheses, Multiplication, Division, Addition, Subtraction
- b) Exponential, Parentheses, Division, Multiplication, Addition, Subtraction
- c) Parentheses, Exponential, Multiplication, Addition, Division, Subtraction
- d) Parentheses, Exponential, Multiplication, Division, Addition, Subtraction**

20. What will be the output of the following Python code?

```
l=[1, 0, 2, 0, 'hello', '', []]  
list(filter(bool, l))
```

- a) [1, 0, 2, 'hello', '', []]
- b) Error
- c) [1, 2, 'hello']
- d) [1, 0, 2, 0, 'hello', '', []]

23. The following python program can work with ____ parameters.

```
def f(x):  
    def f1(*args, **kwargs):  
        print("Sanfoundry")  
        return x(*args, **kwargs)  
    return f1
```

- a) any number of
- b) 0
- c) 1
- d) 2

31. What will be the output of the following Python code snippet?

```
for i in [1, 2, 3, 4][::-1]:  
    print(i, end=' ')
```

- a) 4 3 2 1
- b) error
- c) 1 2 3 4
- d) none of the mentioned

24. What will be the output of the following Python function?

```
min(max(False, -3, -4), 2, 7)
```

- a) -4
- b) -3
- c) 2
- d) False

49. What will be the output of the following Python program?

```
i = 0  
while i < 5:  
    print(i)  
    i += 1
```



```
if i == 3:
    break
else:
    print(0)
```

- a) error
- b) 0 1 2 0
- c) 0 1 2**
- d) none of the mentioned

4. Role: Software Engineer Designing an E-commerce Platform

You are implementing a class in Python to model a `Product` object. Each `Product` has a `name`, `price`, and `category`. You want to ensure that the price of the product cannot be negative. Which OOP concept will you use to implement this?

- A. Inheritance
- B. Polymorphism
- C. Encapsulation
- D. Composition

Correct Answer: C

7. Role: Web Developer Building a Customer Feedback Form

You are writing a Python script to handle customer feedback submissions. If a customer leaves the feedback field empty, you want to raise a custom exception `EmptyFeedbackError`. Which statement should you use to raise this exception?

A.

```
raise Exception(EmptyFeedbackError)
```

B.

```
raise EmptyFeedbackError
```

C.

```
raise empty_feedback_error
```

D.

```
EmptyFeedbackError()
```

Correct Answer: B

12. Role: Machine Learning Engineer Handling Large Datasets

You are working with a large dataset and want to iterate over the data one row at a time to save memory. Which Python feature allows you to create a memory-efficient iterator?

- A. List comprehensions
- B. Generator expressions
- C. Lambda functions
- D. Nested functions

Correct Answer: B

2. Role: Data Scientist Aggregating Customer Data

You are analyzing customer purchases, and you want to calculate the total amount spent by each customer. Which pandas function will you use to group the data by customer ID and calculate the sum of their purchases?

- A. `df.groupby('customer_id').sum()`
- B. `df.groupby('customer_id').mean()`
- C. `df.filter('customer_id').sum()`
- D. `df.agg('customer_id').total()`

Correct Answer: A

6. Role: Marketing Analyst Filtering Campaign Data

You have a dataset containing marketing campaign data. You want to filter the dataset to include only rows where the campaign was marked as successful. Which pandas function will allow you to filter based on this condition?

- A. `df[df['status'] == 'successful']`
- B. `df.filter('successful')`
- C. `df.query('status == "successful"')`
- D. `df[df['status']].filter('successful')`

Correct Answer: A

8. Role: Business Analyst Sorting Product Sales

You have a dataset of products and their sales figures. You want to sort the dataset in descending order of sales to identify the best-selling products. Which method will you use?

- A. `df.sort_values(by='sales', ascending=False)`
- B. `df.sort(by='sales', ascending=True)`
- C. `df.order(by='sales', reverse=True)`
- D. `df.sort(by='sales', descending=True)`

Correct Answer: A

11. Role: Data Scientist Pivoting a DataFrame

You have a dataset of product sales where each row represents a transaction. You want to pivot the data so that each row represents a product, and each column represents a region with sales totals. Which method will help you achieve this?

- A. `df.pivot_table(index='product', columns='region', values='sales', aggfunc='sum')`
- B. `df.groupby('product', 'region').sum()`
- C. `df.pivot(index='product', columns='region', values='sales')`
- D. `df.resample('product', 'region').sum()`

Correct Answer: A

Question 1: DataFrame Column Selection

You have the following DataFrame:

Name	Age	Salary	Department
Alice	25	50000	HR
Bob	30	70000	Finance
Charlie	35	60000	Marketing
David	40	80000	IT

Question:

Which command will return only the "Name" and "Salary" columns from the DataFrame?

- A) `df[['Name', 'Salary']]`
 - B) `df.loc[:, ['Name', 'Salary']]`
 - C) `df.iloc[:, [0, 2]]`
 - D) All of the above
-

Question 2: Filtering Rows

You are given the following DataFrame:

ID	Product	Quantity	Price
1	Laptop	10	800
2	Phone	50	300
3	Tablet	20	400
4	Monitor	15	200

Question:

Which command will return rows where the price is greater than 300?

- A) `df[df['Price'] > 300]`
- B) `df.loc[df['Price'] > 300]`
- C) `df.iloc[df['Price'] > 300]`
- D) Both A and B

Question 3: GroupBy and Aggregation

Consider the following DataFrame:

Employee	Department	Salary
Alice	HR	50000
Bob	IT	60000
Charlie	HR	70000
David	IT	80000

Question:

Which command will calculate the average salary for each department?

- A) `df.groupby('Department')['Salary'].mean()`
- B) `df.groupby('Department').agg({'Salary': 'mean'})`
- C) `df.groupby('Department').mean('Salary')`
- D) Both A and B

4. Role: Data Scientist Handling Missing Data

You need to replace all missing values in a dataset with the median of their respective columns. Which pandas method will allow you to achieve this?

- A. `df.fillna(df.median())`
- B. `df.fillna(df.mean())`
- C. `df.replace(df.median())`
- D. `df.dropna(df.median())`

Correct Answer: A

5. Role: Data Engineer Merging Data from Multiple Sources

You have two datasets: one containing customer information and another containing transaction details. Both datasets have a `customer_id` column. You want to merge these datasets based on this column. Which pandas function will you use?

- A. `pd.merge(df1, df2, on='customer_id')`
- B. `pd.concat([df1, df2])`
- C. `pd.join(df1, df2)`
- D. `pd.union(df1, df2, on='customer_id')`

Correct Answer: A

Question 7: Sorting Data

You have the following DataFrame:

Name	Age	Salary
Alice	25	50000
Bob	30	60000
Charlie	35	70000
David	40	80000

Question:

Which command will sort the DataFrame by "Age" in descending order?

- A) `df.sort_values('Age', ascending=False)`
- B) `df.sort('Age', ascending=False)`
- C) `df.sort_by('Age', order='desc')`
- D) `df.sort_values(by='Age', ascending=True)`

HARD

1. **What is the purpose of `__new__` in Python?**

- A) Returns a class method
- B) Creates a new instance of a class
- C) Initializes an existing instance
- D) Defines a new method

Answer: B) Creates a new instance of a class

2. **How can you ensure a method is always called when an instance is created?**

- A) Use `__del__` method
- B) Use `__call__` method
- C) Use `__new__` method
- D) Use `__init__` method

Answer: D) Use `__init__` method

3. **What is the purpose of `__slots__` in a Python class?**

- A) To define class methods
- B) To initialize class variables
- C) To restrict attribute creation
- D) To create a new instance

Answer: C) To restrict attribute creation

4. **How do you create a custom iterator in Python?**

- A) Implement `__iter__()` and `__next__()` methods
- B) Implement `__next__()` only
- C) Implement `__iter__()` only
- D) Implement `__getitem__()` method

Answer: A) Implement `__iter__()` and `__next__()` methods

5. **What does the `functools.lru_cache` decorator do?**

- A) Caches function results to improve performance
- B) Logs function calls
- C) Limits the number of function arguments
- D) Checks for infinite recursion

Answer: A) Caches function results to improve performance

6. **How do you create a metaclass in Python?**

- A) By defining a class without methods
- B) By using `metaclass` keyword
- C) By inheriting from `type`

D) By inheriting from `object`

Answer: C) By inheriting from `type`

7. **What is the difference between `deepcopy` and `copy` in Python?**

A) `deepcopy` creates a shallow copy, `copy` creates a new object

B) Both are the same

C) `deepcopy` creates a new object with a new memory location, `copy` creates a shallow copy

D) `deepcopy` and `copy` have the same functionality

Answer: C) `deepcopy` creates a new object with a new memory location, `copy` creates a shallow copy

8. **How do you implement method chaining in Python?**

A) By using lambda functions

B) By returning `self` from methods

C) By using decorators

D) By using global variables

Answer: B) By returning `self` from methods

9. **What does the `__contains__` method do in Python?**

A) Initializes the class

B) Creates a new instance

C) Checks membership using the `in` keyword

D) Defines a new class attribute

Answer: C) Checks membership using the `in` keyword

10. **What is a descriptor in Python?**

A) An object that customizes attribute access

B) An object that manages memory

C) An object that defines a class

D) An object that stores data

Answer: A) An object that customizes attribute access

11. **How do you use `contextlib` to manage resources?**

A) By defining a context manager with `@contextmanager`

B) By implementing `__enter__` and `__exit__` methods

C) By using `with` statements directly

D) By using `try...finally` blocks

Answer: A) By defining a context manager with `@contextmanager`

12. What is the use of `__del__` method in Python?

- A) To manage class attributes
- B) To define cleanup actions for an object
- C) To initialize an object
- D) To handle exceptions

Answer: B) To define cleanup actions for an object

13. How do you ensure a function is called only once in Python?

- A) Use a global variable
- B) Use `functools.lru_cache` or a custom decorator
- C) Use `__init__` method
- D) Use `threading.Lock`

Answer: B) Use `functools.lru_cache` or a custom decorator

14. What is the purpose of `__call__` method in Python?

- A) Initializes an object
- B) Creates a new instance
- C) Allows an instance of a class to be called as a function
- D) Defines a class method

Answer: C) Allows an instance of a class to be called as a function

15. How do you create a proxy object in Python?

- A) By inheriting from `object`
- B) By using `__call__` method
- C) By using `proxy` module
- D) By implementing `__getattr__` and `__setattr__` methods

Answer: D) By implementing `__getattr__` and `__setattr__` methods

16. How do you handle cyclic references in Python?

- A) Using manual memory management
- B) Using the garbage collector
- C) By avoiding object references
- D) Using `__del__` method

Answer: B) Using the garbage collector

17. What is the purpose of `__prepare__` method in a metaclass?

- A) To create new instances
- B) To initialize a class
- C) To define class attributes
- D) To prepare the class namespace before creating the class

Answer: D) To prepare the class namespace before creating the class

18. How do you implement multiple inheritance in Python?

- A) By using `super()` method
- B) By using mixins
- C) By inheriting from multiple base classes
- D) By using `__init__` method

Answer: C) By inheriting from multiple base classes

19. What is the `inspect` module used for in Python?

- A) To handle exceptions
- B) To manipulate strings
- C) To get information about live objects, including functions and classes
- D) To manage file operations

Answer: C) To get information about live objects, including functions and classes

20. How can you implement a singleton pattern in Python?

- A) By using a decorator
- B) By using a class variable to store the instance
- C) By using `__new__` method
- D) By using a metaclass

Answer: B) By using a class variable to store the instance

30. What is the order of namespaces in which Python looks for an identifier?

- a) Python first searches the built-in namespace, then the global namespace and finally the local namespace
- b) Python first searches the built-in namespace, then the local namespace and finally the global namespace
- c) Python first searches the local namespace, then the global namespace and finally the built-in namespace
- d) Python first searches the global namespace, then the local namespace and finally the built-in namespace

42. What will be the value of 'result' in following Python program?

```
list1 = [1,2,3,4]
list2 = [2,4,5,6]
list3 = [2,6,7,8]
result = list()
result.extend(i for i in list1 if i not in (list2+list3) and i not in result)
result.extend(i for i in list2 if i not in (list1+list3) and i not in result)
```

```
result.extend(i for i in list3 if i not in (list1+list2) and i not in result)
```

a) [1, 3, 5, 7, 8]

b) [1, 7, 8]

c) [1, 2, 4, 7, 8]

d) error

62. What will be the output of the following Python code?

```
def foo():  
    try:  
        return 1  
    finally:  
        return 2  
k = foo()  
print(k)
```

a) error, there is more than one return statement in a single try-finally block

b) 3

c) 2

d) 1

3. Role: Financial Analyst Summarizing Monthly Reports

You are working with a large financial dataset containing daily transactions. To summarize the data, you want to convert the daily data into monthly totals. Which pandas method allows you to resample the data by month?

A. `df.resample('M').sum()`

B. `df.groupby('M').sum()`

C. `df.filter('M').sum()`

D. `df.reindex('M').sum()`

Correct Answer: A

9. Role: Financial Analyst Calculating Moving Averages

You are analyzing stock prices and want to calculate the rolling 7-day moving average of stock prices. Which pandas function will allow you to do this?

A. `df['price'].rolling(7).mean()`

B. `df['price'].groupby(7).mean()`

C. `df['price'].resample(7).mean()`

D. `df['price'].expanding(7).mean()`

Correct Answer: A

Match List I with List II

List I		List II	
A.	DataFrame.groupby()	I.	To select a single column or subset of columns
B.	DataFrame.describe()	II.	To calculate summary statistics for numerical columns
C.	DataFrame.loc[]	III.	To split the data into groups based on some criteria
D.	DataFrame.iloc[]	IV.	To select rows based on integer location
E.	DataFrame.merge(['column_name'])	V.	To merge DataFrame or Series objects with a database-style join
F.	DataFrame.merge()	VI.	To filter rows based on a condition

Choose the correct answer from the options given below:

1. A - II, B - III, C - I, D - IV, E - VI, F - V

2. A - VI, B - I, C - II, D - IV, E - III, F - V

3. A - III, B - II, C - IV, D - VI, E - I, F - V

4. A - III, B - II, C - VI, D - IV, E - I, F - V

Ans : 4